

SOLUTION Midterm II Part A

Shafayat Abrar

Solution from the answer sheet of Hyder, Afsah.

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Q2

$$B = \frac{\mu_0 I}{2\pi r} \hat{\phi}$$


$$\hat{\phi} = -\hat{x} \sin\phi + \hat{y} \cos\phi$$

$$r = \sqrt{x^2 + y^2}$$

$$ds = dy dz \hat{x}$$

$$B \cdot ds = \frac{\mu_0 I}{2\pi \sqrt{x^2 + y^2}} (-\hat{x} \sin\phi + \hat{y} \cos\phi) \cdot dy dz \hat{x}$$

$$B \cdot ds = -\frac{\mu_0 I}{2\pi \sqrt{x^2 + y^2}} \sin\phi dy dz$$

$\therefore \sin\phi = \frac{y}{\sqrt{x^2 + y^2}}$

$$B \cdot ds = -\frac{\mu_0 I y}{2\pi (x^2 + y^2)} dy dz \quad \text{at } x=b$$

$$\oint B \cdot ds = -\frac{\mu_0 I}{2\pi} \left[\int_{-b}^b dz \right] \left[\int_0^b \frac{y}{b^2 + y^2} dy \right]$$

$$= -\frac{\mu_0 I b}{\pi} \int_0^b \frac{y}{b^2 + y^2} dy$$

$$\int_0^b \frac{y}{b^2 + y^2} dy = \frac{1}{2} \ln(b^2 + y^2) \Big|_0^b$$

$$\frac{1}{2} \ln(2b^2) - \frac{1}{2} \ln(b^2)$$

$$\frac{1}{2} \ln\left(\frac{2b^2}{b^2}\right) = \frac{1}{2} \ln 2$$

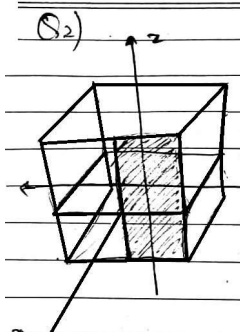
$$\frac{1}{2} \ln(2) \times \left(\frac{-\mu_0 I b}{\pi} \right)$$

$$\boxed{\phi = -\frac{\mu_0 \ln(2) b I}{2\pi}} \quad \text{proved!}$$

I am glad that several students were able to solve this problem correctly. regards --sa

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Q2)



$$\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{\phi} = \frac{\mu_0 I}{2\pi \sqrt{x^2 + y^2}} \hat{\phi}$$

$$\hat{\phi} = \frac{\mu_0 I}{2\pi \sqrt{x^2 + y^2}} (-\sin\phi \hat{x} + \cos\phi \hat{y})$$

$$= \frac{\mu_0 I}{2\pi \sqrt{x^2 + y^2}} \left(\frac{-y}{\sqrt{x^2 + y^2}} \hat{x} + \frac{x}{\sqrt{x^2 + y^2}} \hat{y} \right)$$

$$\vec{B} = \frac{-\mu_0 I y}{2\pi (x^2 + y^2)} \hat{x} + \frac{\mu_0 I x}{2\pi (x^2 + y^2)} \hat{y}$$

$$\Phi = \int \vec{B} \cdot d\vec{s}$$

this expression will be dropped as no flux is passing through this surface in the $\hat{y}$ direction ($\hat{y} \perp \hat{x}$)

$$= \int_{-b}^b \int_{-b}^b \frac{\mu_0 I}{2\pi} \left(\frac{-y}{x^2 + y^2} \hat{x} + \frac{x}{x^2 + y^2} \hat{y} \right) \cdot dz dy \hat{x}$$

$$= \frac{\mu_0 I}{2\pi} \int_{-b}^b \int_{-b}^b \frac{-y}{x^2 + y^2} dy$$

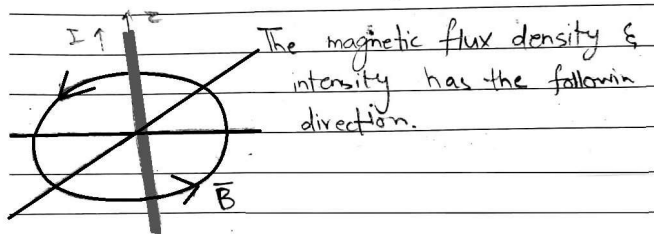
$$= \frac{\mu_0 I}{2\pi} (b - (-b)) \left(-\frac{1}{2} \ln(y^2 + x^2) \right) \Big|_{-b}^b$$

$$= -\frac{\mu_0 I}{2\pi} 2b \cdot \frac{1}{2} [\ln(b^2 + x^2) - \ln(x^2)]$$

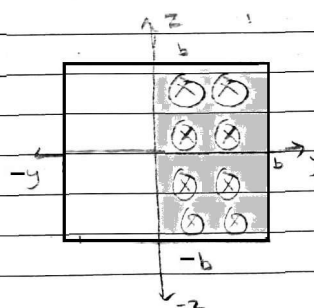
$$= -\frac{\mu_0 I b}{2\pi} \ln\left(\frac{b^2 + x^2}{x^2}\right)$$

$$= -\frac{\mu_0 I b}{2\pi} \ln\left(\frac{b^2}{x^2} + 1\right) \quad \therefore x=b$$

$$= -\frac{\mu_0 I b}{2\pi} \ln\left(\frac{b^2}{b^2} + 1\right) = -\frac{\mu_0 I b}{2\pi} \ln(2)$$



Q2 Solution borrowed from the answer sheet of Rizvi, Syeda Ailiya.



If we look at y-z plane view of the surface we are finding the flux for, we see it is into the paper i.e opposite to the direction of the surface's normal vector. Hence the ϕ has a negative sign with it.

$$\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$$

$$\nabla \times A = \frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{R} & \hat{\theta} & \hat{\phi} \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_R & R A_\theta & (R \sin \theta) A_\phi \end{vmatrix}$$

$$Q_3 \quad B = \nabla \times A$$

$$\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$$

$$A = \frac{\mu_0 I k}{R^2} (\hat{R} \cos \theta - \hat{\theta} \sin \theta)$$

$$\frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{R} & \hat{\theta} & \hat{\phi} \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ \frac{\mu_0 I k \cos \theta}{R^2} & -\frac{\mu_0 I k \sin \theta}{R} & 0 \end{vmatrix}$$

$$\hat{\phi} \frac{1}{R} \left[\frac{\mu_0 I k \sin \theta}{R^2} + \frac{\mu_0 I k \sin \theta}{R^2} \right]$$

$$B = \frac{2 \mu_0 I k \sin \theta}{R^3} \hat{\phi}$$

Q3. Converting from cylindrical to spherical,

$$A = \frac{\mu_0 I k}{R^2} (\hat{R} \cos \theta - \hat{\theta} \sin \theta)$$

$$B = \nabla \times A$$

$$B = \hat{R} \frac{1}{R \sin \theta} \left(\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial}{\partial \phi} A_\theta \right) + \hat{\theta} \frac{1}{R \sin \theta} \left(\frac{\partial}{\partial \phi} A_R - \frac{\partial}{\partial R} (R A_\phi) \right) + \hat{\phi} \frac{1}{R} \left(\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial}{\partial \theta} A_R \right)$$

$$B = \hat{\phi} \frac{1}{R} \left(\frac{\partial}{\partial R} \left(\frac{\mu_0 I k \sin \theta}{R^2} \right) - \frac{\partial}{\partial \theta} \left(\frac{\mu_0 I k \cos \theta}{R^2} \right) \right)$$

$$= \hat{\phi} \frac{1}{R} \left(\frac{\mu_0 I k \sin \theta}{R^2} - \left(\frac{\mu_0 I k (-\sin \theta)}{R^2} \right) \right)$$

$$= \hat{\phi} \frac{1}{R} \left(\frac{\mu_0 I k \sin \theta}{R^2} + \frac{\mu_0 I k \sin \theta}{R^2} \right)$$

$$B = \hat{\phi} \frac{1}{R} \left(\frac{2 \mu_0 I k \sin \theta}{R^2} \right)$$

Borrowed from the answer sheet of Munim Ul Haq

$$\vec{B} = 12x\hat{x} + 25y\hat{y} + cz\hat{z}$$

As divergence of $\vec{B}$ is zero.

$$\nabla \cdot \vec{B} = 0$$

$$0 = \frac{\partial}{\partial x} (12x) + \frac{\partial}{\partial y} (25y) + \frac{\partial}{\partial z} (cz)$$

$$0 = 12 + 25 + c$$

$$c = -37$$

$$\text{So } \vec{B} = 12x\hat{x} + 25y\hat{y} - 37z\hat{z}$$

Borrowed from the work of Masroor, Shameer

Question 1.

We know that

$$\nabla \cdot \vec{B} = 0$$

$$\text{as } \vec{B} = 12x\hat{x} + 25y\hat{y} + cz\hat{z}$$

$$\nabla = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

$$\nabla \cdot \vec{B} = 12 + 25 + c = 0$$

$$\boxed{c = -37}$$

$$\vec{B} = 12x\hat{x} + 25y\hat{y} - 37z\hat{z} \text{ Wb/m}^2$$

Q4 $f(x,y) = 2x - y - 1$
 $\hat{n} = \frac{2\hat{x} - \hat{y}}{\sqrt{5}}$

Borrowed from the work of Hyder, Afsah.

a) $\mu_1 = 4\mu_0$, $\mu_r = 4$, $\mu_2 = 1 + \mu_m$
 $(\mu_m = 3)$

$M_1 = I_m H$

$M_1 = 3(3\hat{x} - 2\hat{y} + 2\hat{z})$

$M_1 = 9\hat{x} - 6\hat{y} + 6\hat{z}$ A/m

$B = \mu_0 (3\hat{x} - 2\hat{y} + 2\hat{z})$
 $\mu_0 = 4\pi \times 10^{-7}$

$B_1 = \mu_0 (12\hat{x} - 8\hat{y} + 8\hat{z})$

$B_1 = 15.08\hat{x} - 10.05\hat{y} + 10.05\hat{z}$

b) No current density on surface, $J=0$

$B_{1n} = B_{2n} \Rightarrow \mu_1 H_{1n} = \mu_2 H_{2n}$

$H_m = (H_1 \cdot \hat{n}) \hat{n}$

$= \left((3, -2, 2) \cdot \frac{(2, -1, 0)}{\sqrt{5}} \right) \left(\frac{(2, -1, 0)}{\sqrt{5}} \right)$

$= \frac{6+2}{\sqrt{5}} \left(\frac{(2, -1, 0)}{\sqrt{5}} \right) = \frac{8}{5} (2, -1, 0)$

$H_m = (3.2, -1.6, 0)$
 $H_m = 3.2\hat{x} - 1.6\hat{y}$

$H_{2n} = \frac{4\mu_0}{3\mu_0} (H_m)$

$H_{2n} = \frac{4}{3} (3.2\hat{x} - 1.6\hat{y})$

$H_{2n} = 4.267\hat{x} - 2.133\hat{y}$

$H_t = H_{1t} + H_{1n}$

$H_{1t} = (3, -2, 2) - (3.2, -1.6, 0)$

$= (-0.2, -0.4, 2)$

$H_{1t} = 0.2\hat{x} - 0.4\hat{y} + 2\hat{z}$

$H_{1t} = H_{2t}$ Since $J=0$

So $H_2 = H_{2n} + H_{2t}$

$= \frac{4}{3} (3.2, -1.6, 0) + (-0.2, -0.4, 2)$

$H_2 = 4.067\hat{x} - 2.533\hat{y} + 2\hat{z}$

$B_2 = 3\mu_0 (H_2)$

$B_2 = \mu_0 (12.2\hat{x} - 7.599\hat{y} + 6\hat{z})$

Q4) $H_1 = 3\hat{x} - 2\hat{y} + 2\hat{z}$ A/m

$\mu_1 = 4\mu_0$

$\mu_2 = 3\mu_0$

region 1: $2x - y - 1 \leq 0$ $f(x,y) = 2x - y - 1$

$\nabla f = 2\hat{x} - \hat{y}$

$\hat{n} = \frac{\nabla f}{|\nabla f|}$
 $|\nabla f| = \sqrt{2^2 + 1^2} = \sqrt{5}$

Borrowed from the work of Rizvi, Syeda Ailiya

a) $M_1 = \chi_{m1} H_1$

$= (\mu_{r1} - 1) H_1$

$= (4-1)(3\hat{x} - 2\hat{y} + 2\hat{z})$

$= 3\langle 3, -2, 2 \rangle = \langle 9, -6, 6 \rangle = 9\hat{x} - 6\hat{y} + 6\hat{z}$

$B_1 = \mu_1 H_1$

$= 4\mu_0 \langle 3, -2, 2 \rangle$

$= 4(4\pi \times 10^{-7}) \langle 3, -2, 2 \rangle$

$= 5.0265 \times 10^{-6} \langle 3, -2, 2 \rangle$

$= 1.5079 \times 10^{-5} \hat{x} - 1.0053 \times 10^{-5} \hat{y} + 1.0053 \times 10^{-5} \hat{z}$ Wb/m²

b) $H_{1n} = (H_1 \cdot \hat{n}) \hat{n}$
 $= \left(\langle 3, -2, 2 \rangle \cdot \frac{\langle 2, -1, 0 \rangle}{\sqrt{5}} \right) \frac{\langle 2, -1, 0 \rangle}{\sqrt{5}}$

$= \frac{6+2+0}{5} \langle 2, -1, 0 \rangle$

$H_{1n} = \frac{8}{5} \langle 2, -1, 0 \rangle = \frac{16}{5} \hat{x} - 8\hat{y}$

$\therefore H_1 = H_{1n} + H_{1t}$

$3\hat{x} - 2\hat{y} + 2\hat{z} = \frac{16}{5}\hat{x} - 8\hat{y} + H_{1t}$

$H_{1t} = \left(3 - \frac{16}{5} \right) \hat{x} + \left(-2 + \frac{8}{5} \right) \hat{y} + 2\hat{z}$

$= -\frac{1}{5}\hat{x} - \frac{2}{5}\hat{y} + 2\hat{z}$

Acc to boundary conditions we know

$H_{1t} = H_{2t}$

$H_{2t} = \langle -1/5, -2/5, 2 \rangle$

$B_{2n} = B_{1n} \Rightarrow \mu_2 H_{2n} = \mu_1 H_{1n}$

$\Rightarrow H_{2n} = \frac{\mu_1}{\mu_2} H_{1n}$

$\Rightarrow H_{2n} = \frac{4}{3} \cdot \langle 16/5, -8/5, 0 \rangle$

$\Rightarrow H_{2n} = \frac{64}{15} \hat{x} - \frac{32}{15} \hat{y}$

$H_2 = H_{2n} + H_{2t}$

$= \langle \frac{64}{15}, -\frac{32}{15}, 0 \rangle + \langle -1/5, -2/5, 2 \rangle$

$= \langle \frac{61}{15}, \frac{38}{15}, 2 \rangle$ A/m

$B_2 = \mu_2 H_2$

$= 3\mu_0 \langle \frac{61}{15}, \frac{38}{15}, 2 \rangle = \frac{61}{5} \mu_0 \hat{x} + \frac{38}{5} \mu_0 \hat{y} + 6\mu_0 \hat{z}$
Wb/m²

I am so glad that I have students in the class whom I may borrow the solutions from...